Supplementary Table 1 – Characteristics of included studies (n = 31)

	Source (Y)	Sample Size	Birth weight (g) (study group)	Gestation (wks.) (study group)	Age at which outcomes measured (years)	Outcome measure (for example eGFR or BP or KV)	Conclusion (briefly)
1	Downing et al (1992) [15]	Control N=7 (Premature no frusemide therapy and no renal calcifications) Group 2 N=10 (Premature frusemide therapy but no renal calcifications) Group 3 N=10 (Premature frusemide therapy and renal calcifications)	Control (743(84)) Group 2 (806(103)) Group 3 (678(95))	Control (25.3(0.6) Group 2 (26.1(0.5)) Group 3 (26.2(0.4))	Control (1.2(0.01)) Group 2 (1.3 (0.1)) Group 3 (1.2 (0.1))	Creatinine clearance Urinary calcium:creatinine ratio Fractional excretion of sodium Lower tubular reabsorption of phosphate Urine-blood difference in carbon dioxide tension after oral acetazolamide load Proteinuria	• No significant difference in renal function between the control group and group 2 • In group 3, trace proteinuria was present in 4/10 participants • Creatinine clearance in group 3 was significantly lower than in the control group and group 2 • Urinary calcium:creatinine ratios, fractional excretion of sodium and lower tubular re-absorption of phosphate was significantly higher in Group 3 compared to the group 2 and control group • Group 3 had lower urine-blood differences in carbon dioxide tension after oral acetazolamide load when compared to the controls and group 2
2	Jones et al (1997) [16]	Control Group (premature without renal calcifications) N=17 Study Group (premature with renal calcifications) N=11	Control (982 (710–1760)) Study (850 (580–1856))	Control (28 (25–31)) Study (27 (24–31))	4-5	eGFR Renal calcifications/ nephrocalcinosis	• In the study group the median GFR was 61 ml/min/1.73m2 (range 46-79 ml/min/1.73m2) • Five of the 11 children born premature and found to have renal calcifications, still had renal calcifications at age 4-5 years. • Nephrocalcinosis in isolation was not found to be a major predisposing factor to long term renal dysfunction • Four out of the 11 premature babies with nephrocalcinosis had an abnormal GFR at 4-5 years of age • Two children from the collective premature cohort had microalbuminuria at 4-5 years of age

3	Ojala et al (2001) [17]	Control Group N=35 (premature without perinatal indomethacin) Study Group N=31 (premature with perinatal indomethacin exposure)	Control Group (1360 (680–2680)) Study Group (1150 (670–2060))	Control Group (31 (24-32)) Study Group (28 (24-32))	2-4	Serum cystatin C and protein Plasma creatinine, sodium and potassium Urine PCR Urine alpha-1 microglobulin eGFR BP Renal sonography examination	• The control group showed higher mean serum cystatin concentrations than the study group • No difference was found between the groups for mean serum protein, plasma creatinine and sodium, median plasma potassium concentrations, urine protein:creatinine ratio and urine calcium:creatinine ratio. • No statistical difference was found in GFR or renal structural abnormalities between the two groups • Nil significant difference in urine PCR between groups • Umbilical artery catheter use, frusemide treatment and assisted ventilation may correlate with long term renal structural and functional abnormalities
4	Abitbol et al (2003) [18]	N=20	686 (133)	25 (2)	3.1-18.3	Proteinuria Kidney function/kidney failure (eGFR, sCr) Renal size and mass BP Growth (height, weight and BMI) Nephrocalcinosis	• Significant age difference was found between the normal GFR and low GFR groups, the age of outcome measure was 5.7+/-2.2 and 9.9+/-5.6 years respectively. • BP was not significantly different between study participants and population norms • Urine PCR was available for 10 out of 20 of the participants. Participants with low GFR (n=3) had significantly higher urine protein/creatinine ratios compared to the participants with normal GFR (n=7) at follow up.
5	Rodriguez-Soriano et al (2005) [19]	Control Group N=43 Study Group N=40 (premature and weighing <1000g at birth)	845 (540-1000)	27.6 (23-35)	Control Group (8.5 (1.8)) Study Group (8.6(1.8))	BP Renal length and volume Plasma creatinine Estimated creatinine clearance TmP/GFR TRP Urinary phosphate excretion Urinary calcium excretion eGFR	• No significant difference in BP, microalbuminuria, and renal length and volume was found between the study and control groups • Plasma creatinine, urinary calcium excretion and urinary phosphate excretion were significantly higher in the study group than the control group • The study group had significantly lower estimated creatinine clearance and TmP/GFR compared to controls • No significant differences were observed in microalbuminuria values, but five study subjects (12.5%) presented values above the upper limit of normal.
6	Porter et al (2006) [20]	Control Group N=14 (without nephrocalcinosis – matched for birth weight, gestational age and sex)	Control Group (1210 (670-1870)	Control Group (28.5 (25-31)) Study Group (27.5 (25-31)	Control Group (7.21 (6.38-7.68)) Study Group (6.69 (5.81-7.09))	Early morning urine osmolality Urine ACR and PCR Creatinine:phosphate ratio Creatinine:calcium ratio Beta microglobulin	• No significant differences in GFR or urinary concentrating capacity between the groups • 75% of patients who underwent renal ultrasound were found to have resolved nephrocalcinosis by a median age of 6.75 years • There was evidence of hypercalciuria in both the control and study groups suggesting prematurity may be a risk factor

		Study Group N=14 (very low birth weight, premature and with nephrocalcinosis)	Study Group (1180 (565-1880)			UEC eGFR TmP/GFR Renal length Nephrocalcinosis	• No evidence suggested that nephrocalcinosis is associated with long term renal dysfunction • Nil significant difference in urine ACR and PCR in controls and patients
7	Kist-van Holthe et al (2007) [21]	Control N=32 (Premature without Nephrocalcinosis) Study Group N=42 (Premature with neonatal Nephrocalcinosis)	Control Group (1353 (337)) Study Group (1148 (394))	Control Group (29.8 (1.6)) Study Group (28.9 (2.3))	Control Group (7.5 (1.0)) Study Group (7.4 (1.0))	BP eGFR Tubular function Nephrocalcinosis Kidney length	• There was no difference in the BP between the two groups • Blood pressure in both groups was found to be higher than expected for otherwise healthy children • The study group was found to have significantly more chronic renal insufficiency when compared to healthy children. This was not the case for the control group. • Tubular function, urine albumin, kidney length and GFR was not significantly different between the study and control groups • Tubular phosphate reabsorption, plasma bicarbonate, and early-morning urine osmolality were significantly lower in both control and study groups when compared to otherwise healthy children
8	Keijzer-Veen et al (2007) [22]	Control (Term) N=30 Group 1 (Premature SGA) N=23 Group 2 (Premature AGA) N=29	Control (3632 (40.2)) Group 1 (859 (126)) Group 2 ((1489 (257))	Control (40.2 (1.3)) Group 1 (30.6 (1.0)) Group 2 (29.5 (1.4))	20	eGFR Serum urea sCr Serum electrolytes ERPF BP Albuminuria Kidney length and volume	• Height, weight, kidney length and volume, GFR, and ERPF were significantly lower in the SGA group than in controls. After adjustment for body surface area, GFR did not differ significantly among groups. • There was increased BP in premature compared to controls • Two participants from group 1 had microalbuminuria
9	Finken et al (2008) [23]	Control N=328 (premature and did not receive betamethasone) Study Group N=84	Control (1319 (337)) Study Group (1348 (275))	Control (29.7 (1.5)) Study Group (29.8 (1.5))	19	Body composition Insulin resistance Serum lipid profile BP eGFR	• eGFR was found to be lower in 19-year-old born premature who received antenatal betamethasone • This difference was clinically irrelevant at age 19, however the decreased eGFR may increase the risk of chronic kidney disease long term

		(premature and received betamethasone)					
10	Rakow et al (2008) [24]	Control (term AGA) N=37 Group 1 (premature) N=39 Group 2 (term SGA) N=29	Control (3485 (502)) Group 1 (954 (203)) Group 2 (2436 (331))	Control (39.6 (1.0)) Group 1 (26.6 (2.0)) Group 2 (39.3 (1.4))	Control (9.8 (0.2)) Group 1 (9.6 (0.3)) Group 2 (9.8 (0.3))	eGFR Kidney volume sCr Serum Cystatin C BP Urinary ACR, Immunoglobulin G, alpha-1 microglobulin, N-acetylglucosamine	• Estimated glomerular filtration rate (eGFR) and urinary protein patterns were similar between the groups. • Kidney volume was smaller in the premature group than in the controls, but the difference was not significant when adjusted for body surface area, gender and age • No significant differences were found in renal function, urine ACR, renal volume or blood pressure between the three groups at school age.
11	Bacchetta et al (2009) [25]	Control (Premature normotrophic children) N=11 Group 1 (Premature EUGR) N=16 Group 2 (Premature IUGR) N=23	Control (1039 (278)) Group 1 (845(146)) Group 2 (773 (155))	Control (27.1 (1.8)) Group 1 (26.2(1.8)) Group 2 (28.2 (1.8))	Control (6.8 (0.9)) Group 1 (7.9 (1.3)) Group 2 (7.8 (1.3))	BP eGFR Microalbuminuria Urine calcium-creatinine ratio Kidney size	• Children in groups 1 and 2 had decreased GFR compared to controls • Nil significant difference in blood pressure was found between the three groups • EUGR was concluded as a risk factor for long term renal impairment in premature children • Two children out of the entire cohort had moderate microalbuminuria
12	Zaffanello et al (2010) [26]	Group 1 (VLBW) N=43 Group 2 (ELBW) N=26	Group 1 (1315 (1248–1352)) Group 2 (850 (775–883))	Group 1 (30.1 (29.9–31.3)) Group 2 (27.0 (26.3–27.7))	Group 1 (5.4 (5.2–6.1)) Group 2 (5.3 (5.2–6.3))	Plasma creatinine concentration Plasma Cystatin C eGFR Plasma renin Urinary alpha 1-microglobulin Albuminuria Total kidney volume BP	• Renal function parameters (i.e., estimated glomerular filtration rate and albuminuria) did not differ between the two groups of children. • Systolic and diastolic blood pressures and did not differ between the two birth- weight categories.

13	Chan et al (2010) [27]	Control (Term AGA) N=25 Group 1 (Premature SGA) N=14 Group 2 (Premature AGA) N=25 Group 3 (Term SGA) N=7	Control (3302 (3105-3690)) Group 1 (980(768-1038)) Group 2 (1635 (991-1850)) Group 3 (2750 (2430-2870))	Control (40 (38.5-41.0)) Group 1 (31(28.8-31.0)) Group 2 (30 (27.5-31.0)) Group 3 (39 (38.0-40.0)	Control (13.6 (12.54-14.78)) Group 1 (13.5 (12.48-13.97)) Group 2 (14.1 (13.66-15.03)) Group 3 (13.6 (12.35-14.83))	BP Augmentation index eGFR following protein load Plasma glucose Serum insulin levels	• Nil difference in GFR prior to giving participants a protein load. • SGA had higher SBP and lower GFR following protein load than AGA. There was no effect of prematurity on SBP or GFR
14	Giapros et al (2011) [28]	Control Group (Premature without nephrocalcinosis) N=44 Study Group (premature with nephrocalcinosis) N=63	Control (1651 (430)) Study (1615 (480))	Control (31.9 (2.2)) Study (31.8 (3))	0.25, 0.5, 1, 2	sCr eGFR Fractional excretion of sodium, potassium, phosphate, magnesium and uric acid Kidney length	• sCr and eGFR did not differ between the groups at any time point • The NC group had a shorter KL up to 12 months of life (left kidney) or 24 months (right kidney) • Nephrocalcinosis in premature infants was associated with renal tubular dysfunction and shorter kidney length in the first year of life
15	Carballo-Magdaleno et al (2011) [29]	Control N=30 Group 1 N=30 (premature and no steroids) Group 2 N=30 (premature with steroids)	Control (3088 (177)) Group 1 (1669 (426)) Group 2 (1501 (410))	Control (38.6 (1.0)) Group 1 (31.9 (2.3)) Group 2 (31.3 (1.7))	Control (2.0 (0.4)) Group 1 (1.8 (0.5)) Group 2 (1.8 (0.4))	Renal volume eGFR Cystatin C BP	• Groups 1 and 2, when compared to the controls had higher BP, cystatin C levels and GFR • No significant difference in these parameters was found between groups 1 and 2 • Concluded that prematurity (independent of antenatal steroids) was associated with higher blood pressure levels, cystatin C levels and glomerular filtration rates in infants aged 12-36 months

16	Starzec et al (2016) [30]	Control (term) N=36 Study Group (ELBW) N=64	Control (3570 (3395–3880)) Study (875 (750–960))	Control (40 (39–40)) Study (27 (25–28))	7, 11	Serum cystatin C levels sCr BUN eGFR Kidney length and width	• Renal ultrasound examination revealed a significantly smaller renal volume in the 7- and 11-year-old ELBW children compared to the term controls • Serum cystatin C levels were significantly higher in ELBW children than in the controls at 7 years of age, and this difference remained statistically significant at 11 years of age • GFR was significantly lower in the extremely low birth weight group compared to term controls when assessed at 11 years
17	Bruel et al (2016) [31]	Control (premature without AKI) N=25 Study Group (premature with AKI) N=49	Control (1034 (853-1348)) Study (815(708-1110))	Control (28(27-30)) Study (28 (26-29.2))	Control (7 (5-8)) Study (7 (5-8))	eGFR BP ACR Albuminuria Microalbuminuria Renal volume	• There was no difference in microalbuminuria or eGFR at the time of follow up • No significant difference in eGFR between groups. • Renal volume was significantly lower in the study group • In the collective premature cohort, 10.8% had microalbuminuria and 23% had diminished eGFR • eGFR was noted to be significantly lower in children with VLBW <1000g • Blood pressure was not significantly difference between both groups
18	Masqood et al (2017) [32]	Control (ELBW with no AKI) N=112 Group 1 (ELBW with AKI stage 1) N=87 Group 2 (ELBW with AKI Stages 2 and 3) N=23	Control (813 (127)) Group 1 (725(140)) Group 2 (664(172))	Control (26.8(2.3)) Group 1 (25.5(2.0)) Group 2 (25.3(1.9))	Control (6.6(2.5)) Group 1 (6.9(2.9)) Group 2 (8.3(3.3))	BP BMI Creatinine BUN eGFR	• No significant difference in the prevalence of CKD between groups, however ELBW was noted to increase the risk of CKD independent of neonatal AKI • There was no difference in growth parameters or prevalence of hypertension or prehypertension among the three groups at their latest follow up (age of outcome measure)
19	Harer et al (2017) [33]	Control (term) Group 1 (premature VLBW with AKI) N=20	Control Group 1 (790(730-1018)) Group 2 (1040(855-1443))	Control Group 1 (25(24-26)) Group 2 (29(27-29))	Control Group 1 (5(4-5)) Group 2 (5(4-6))	eGFR Urine PCR BP Cystatin C Kidney size	• There was a significant difference in the cystatin C values when the premature infants were compared with the term infants. The premature groups combined had higher mean cystatin C values that the term cohort • 65% of premature children with AKI had kidney dysfunction at a median 5 years of age and 14% of premature children with no history of AKI has kidney dysfunction aged 5 years. • Nil significant difference in Urine PCR between groups

		Group 2 (premature VLBW without AKI) N=14					
20	Raaijmakers et al (2017) [34]	Control = 45 (premature and exposed to ibuprofen) Study Group = 48 (premature and not exposed to ibuprofen)	815 (430-1000)	27 (24-33)	0.75, 2, 11	eGFR-Cystatin C Renal length	• There with no significant differences in renal length or eGFR-Cystatin C in young adolescence who experienced neonatal ibuprofen exposure
21	Vollsaeter et al (2018) [35]	Control (Term Born AGA) N=54 Group 1 (Premature-born AGA) N=37 Group 2 (Premature-born SGA) N=20	Control (3701 (3582, 3581)) Group 1 (918 (867, 968)) Group 2 (724 (657, 791))	Control = Term born Group 1 (26.1 (25.7, 26.5) Group 2 (28.0 (27.2, 28.7)	Control (11.7 (11.2-12.0)) Group 1 (11.4 (11.1-11.8)) Group 2 (11.3 (11.0-11.8))	Height Weight Abdominal circumference Triceps and subscapular skin fold thickness BP Plasma creatinine Cystatin C eGFR SDMA	• SDMA levels were significantly higher Group 1 and 2 when compared to the controls • GFR was significantly lower in Groups 1 and 2 compared to the controls • No significant differences in creatinine or cystatin C between the groups • Systolic BP had a significant relationship with fat mass indices but not renal function • Systolic BP did not significantly differ between the groups • Concluded that children from Groups 1 and 2 (especially Group 2) had impaired renal function by 11 years of age (as shown by GFR and SDMA) • Findings suggest being born premature or SGA increases risk of developing kidney disease in the future
22	South et al (2019) [36]	Control (term) N=43 Study group (premature with VLBW) N=96	Control (3458 (451)) Study (1048 (276))	Control (39.7 (1.1)) Study (27.8 (2.6))	Control (14) Study (14)	BP ACR eGFR BMI BUN	• Premature children had significantly higher BP and significantly lower eGFR compared to the term children at 14 years of age • Albuminuria, ACR, creatinine and BUN were not significantly different

23	Crump et al (2019) [37]	Extremely Premature N=8129 Very premature N=43516 Late premature N=155626 Early term N=737412 Full term N=2895746 Post-term N=346186	NA	Extremely Premature (22-27) Very premature (28-33) Late premature (34-36) Early term (37-38) Full term (39-41) Post-term (≥42)	0-43	CKD	• Premature and extremely premature birth were associated with twofold and threefold risks of CKD respectively • Premature birth and CKD were found to have the strongest association at ages 0-9 years (hazard ratio 5.09) and weakened but remained increased at ages >9 years (hazard ratio 1.97 for 10-19 years and 1.34 for 20-43 years)
24	Rakow et al (2019) [38]	Control N=19 Group 1 (Extremely premature and nephrocalcinosis) N=20 Group 2 (extremely premature and no nephrocalcinosis) N=21	Control (3586 (477)) Group 1 (755 (124)) Group 2 (841 (202))	Control (39.7 (1.6)) Group 1 (25.5 (1.2)) Group 2 (25.9 (1.3))	Control (8.1 (2.2)) Group 1 (7.8 (1.0)) Group 2 (7.4 (1.1))	Kidney volume 24-hour ambulatory BP Cystatin C calculated eGFR Plasma creatinine Urinary PCR Urine electrolytes	• Groups 1 and 2 had significantly smaller kidneys compared to the controls • Cystatin C based GFR was significantly lower (however still normal) in groups 1 and 2 when compared to the control group • Nil significant difference between kidney volume and function between Groups 1 and 2 • The control groups had significantly higher plasma creatinine compared to groups 1 and 2 • Urinary PCR and electrolytes were not significantly different between all groups • BP was not significantly different between all groups • Significantly more children from group 1 had a negative evolution of kidney function from the neonatal period to school age

25	Horie et al (2019) [39]	Control Group (Premature and normal eGFR) N=150 Study Group (Premature and low eGFR) N=18	<1000g N=73 1000-1500g N=76 >1500g N=19	<28+0 N=63 ≥28+0 N=105	2-15	sCr eGFR Gestational age Body weight and length at birth Sex Apgar score Use of antimicrobial agents, steroids or indomethacin	• eGFR at 2 years of age was significantly correlated with birthweight and gestational age. This relationship was no longer significant at 3-4 years of age. • Approximately 10.7% of the children had low eGFR without clinical symptoms or abnormal urine examination. These children had high sCr on day 7 after birth and delayed recovery of these levels during the first month after birth.
26	Crump et al (2019) [40]	Extremely Premature N=8324 Very premature N=44373 Late premature N=157342 Early term N=740391 Full term N=2896444 Post-term N=346195	NA	Extremely Premature (22-27) Very premature (28-33) Late premature (34-36) Early term (37-38) Full term (39-41) Post-term (≥42)	0-43	BP	• In this study, premature birth was associated with increased risk of hypertension in early adulthood • Adjusted hazards ratios for new-onset hypertension at 18-29 years of age associated with premature (<37 weeks) and extremely premature (22-27 weeks) were 1.28 and 2.45 respectively when compared to full term birth. At ages 30-43 years the adjusted hazards ratios were 1.25 and 1.68 respectfully for premature and extremely premature birth when compared to full term birth.
27	Kandasamy et al (2020) [41]	Control group (term) N=31 Study Group (premature) N=53	NA	Control = Born at term Study = <28	0.5, 1, 2	TKV eGFR Urine ACR BP	• The study group had a significantly reduced TKV compared to the controls • Both groups had a similar eGFR • No significant difference was found in BP and urine ACR between the groups

28	Sanderson et al (2020) [42]	N=42 (premature)	770 (173.1)	25.7 (1.1)	15 (15, 15.3).	Albuminuria BP Kidney volume	• 33.3% of participants had an elevated BP at 15 years of age • 11.9% of the cohort had microalbuminuria • 14% had a kidney volume below the 10th percentile of normative data • 50% of the sample had at least one kidney abnormality (microalbuminuria, elevate BP and/or kidney hypoplasia
29	Yael et al (2020) [43]	Control group = known paediatric population prevalence rates Study group (VLBW and premature) N = 103	Study Group (1086 (243))	Study Group (29.4 (24-35))	Study Group (11.6 (10-13.3))	BP Urine PCR Urine ACR eGFR	• The prevalence of systolic hypertension was 15.8% and of systolic pre-hypertension was 6.9%. This is compared to the general United States paediatric population which has a 1.6% prevalence rate of systolic hypertension. • Hypertension was associated with a significantly diminished mean birth weight compared to the remainder of the cohort (939.3 grams vs 1111 grams P=0.024) • 103 study participants that were very low birth weight and premature all had normal values of GFR • Prevalence of microalbuminuria was 14.3% while the prevalence of proteinuria was 7.9% at 11.6 years.
30	Staub et al (2020) [44]	Control Group (term) N=82 Study Group (premature) N=51	Control (3250 (555)) Study (1360 (532))	Control (39 + 6 (2 + 5)) Study (31 + 0 (2 + 6))	Control (12.1 (1.20)) Study (12.3 (1.87))	BP sCr Cystatin C eGFR Beta-2 Microglobulin Uromodulin Neutrophil gelatinase-associated lipocalin	• Systolic BP was significantly higher in premature boys compared with term boys, however there was not significant difference in girls • Low birth weight was associated with higher BP in premature boys • In the premature group, maternal hypertension/preeclampsia and adolescent height were associated with higher systolic BP • sCr and neutrophil gelatinase-associated lipocalin were significantly higher in the premature group • There was no significant difference in GFR between groups
31	Askenazi et al (2021) [45]	N=923 (extremely low gestational age)	801.1 (187.9)	24-27	1.83-2.17	eGFR Urine ACR BP	• The prevalence of Stage 2 or 3 AKI was 18.2% • At 22-26 months 16% had an eGFR <90mL/min/1.73m^2, 35.8% had elevated urine ACR, 23% had a SBP > 95th percentile for their age and 40% had a DBP >95th percentile for age

Abbreviations: BP (blood pressure), SBP (systolic blood pressure), DBP (diastolic blood pressure) eGFR (estimated glomerular filtration rate), BMI (body mass index), TmP (tubular maximum reabsorption of phosphate), sCr (serum creatinine), CKD (chronic kidney disease), ACR (albumin creatinine ratio), PCR (protein creatinine ratio), SDMA (Symmetric dimethylarginine), TKV (total kidney volume), IUGR (intrauterine growth retardation), EUGR (extrauterine growth retardation), SGA (small for gestational age), AGA (appropriate for gestational age), BUN (blood urea nitrogen), ERPF (effective renal plasma flow), AKI (acute kidney injury), VLBW (very low birth weight), ELBW (extremely low birth weight)